Summary

Section 9-1: WAN Connectivity

- A WAN traverses a significant distance and usually supports very high data throughput. Although many types of businesses need WANs, they might not need the same kinds of WANs.
- Two categories of WAN connectivity services are broadband, where the cables (whether telephone, coaxial, or fiber) and available bandwidth are shared between multiple customers, and DIA (dedicated Internet access), where the cable or a portion of its available bandwidth is dedicated to a single customer.
- With a direct connection to the cloud, you lease a dedicated line from your ISP to one of your cloud provider's PoPs, or colocation facilities. From there, you pay for the connection to the cloud provider's physical infrastructure and, usually, some kind of data transfer fees.
- SD-WAN (software-defined wide area network) relies on abstracted, centralized control of networking devices to manage network functions across a diverse infrastructure. SD-WAN offers the following benefits: transport agnostic, AAR (application aware routing), active-active load balancing, automatic failover, intent-based management, zero-touch provisioning, and reduced cost compared to services such as leased lines and MPLS.
- 5G (fifth generation) services require minimum theoretical speeds of 1 Gbps and max out at 20 Gbps download and 10 Gbps upload; however, actual speeds vary greatly. 5G relies on three frequency bands called low band, mid-band, and high band.
- Satellite Internet clients can exchange signals with satellites as long as they have a line-of-sight path from an unobstructed view of the sky. To establish a satellite Internet connection, each subscriber must have a fixed dish antenna pointed toward the sky over the equator.

Section 9-2: Routing Protocols

- A router joins two or more networks and passes packets from one network to another. Routers are responsible for determining the next network to which a packet should be forwarded on its way to its destination.
- Routers are often categorized according to their location on a network or the Internet and the routing protocols they use. The various categories include core routers, edge routers, and exterior routers.
- A routing table is a database that holds information about where hosts are located and the most efficient way to reach them. A router relies on its routing table to identify which network a host belongs to and which of the router's interfaces points toward the best next hop to reach that network.
- To determine the best path, routers communicate with each other through routing protocols. Routers rate the reliability and priority of a routing protocol's data based on AD (administrative distance), convergence time, and overhead.
- IGPs (interior gateway protocols) are routing protocols used by core routers and edge routers within autonomous systems. IGPs are often grouped according to the algorithms they use to calculate best paths, including distance-vector routing protocols, link-state routing protocols, and hybrid routing protocols. EGPs (exterior gateway protocols) are routing protocols used by edge routers and exterior routers to distribute data outside of autonomous systems. The only EGP protocol currently in use is BGP.
- Three popular FHRPs (First Hop Redundancy Protocol) used by routers and layer 3 switches to provide a single VIP (virtual IP) address as the default gateway to a network are VRRP (Virtual Router Redundancy Protocol), HSRP (Hot Standby Routing Protocol), and GLBP (Gateway Load Balancing Protocol).

Section 9-3: Securing WAN Connections

- A router's main function is to examine packets and determine where to direct them based on their network layer addressing information. Thanks to a router's ACL (access control list), or access list, routers can also decline to forward certain packets depending on their content.
- A proxy server, or proxy, acts as an intermediary between the external and internal networks, screening all incoming and outgoing traffic. Although proxy servers only provide low-grade security relative to other security devices, they can help prevent an attack on internal network resources such as web servers and web clients.
- A firewall is a specialized device or software that selectively filters or blocks traffic between networks. A firewall might be placed internally, residing between two interconnected private networks. More commonly, the firewall is placed on the edge of the private network, monitoring the connection between a private network and a public network, such as the Internet.

- Cloud security technologies have evolved, with many cloud platforms embedding security services to protect against security challenges unique to the cloud. Most cloud security tools offer granular control, stateful traffic filtering, and predictable default configurations.

Section 9-4: Troubleshooting WAN Connections

- Common issues causing Internet connectivity problems include interference, DNS issues, router misconfiguration, interface errors, and roaming misconfiguration.
- Interface configuration, status, and statistics can all provide helpful information in troubleshooting a network connection problem. To get an overview of all the device's interfaces, enter the command `show interface` (or `sh int`).
- Just as interfaces must be properly configured for network connections to work as expected, misconfigured routing tables also can cause problems for network connections. On a Cisco router, the `show ip route` command lists the router's routing table information.

Key Terms

For definitions of key terms, check the Glossary.

4G (fourth generation)
5G (fifth generation)
AAR (application-aware routing)
ACL (access control list)
AD (administrative distance)
application layer firewall
AS (autonomous system)
asymmetrical
bandwidth speed tester
best path
BGP (Border Gateway Protocol)
broadband
cable broadband
cable modem
cell site
colocation facility
convergence time
core router
CRC (Cyclic Redundancy Check) error
default route
DIA (dedicated Internet access)
direct connection
distance-vector routing protocol
DOCSIS (Data Over Cable Service Interface Specifications)
DSL (digital subscriber line)
DSL modem
dynamic routing
edge router
EGP (exterior gateway protocol)
EIGRP (Enhanced Interior Gateway Routing Protocol)
exterior router
FHRP (First Hop Redundancy Protocol)
gateway of last resort
giant
host-based firewall
hybrid routing protocol
IGP (interior gateway protocol)
implicit deny
IS-IS (Intermediate System to Intermediate System)
leased line
link-state routing protocol
LTE (Long-Term Evolution)
LTE-A (LTE-Advanced)
metric
modem
MPLS (multiprotocol label switching)
network-based firewall
NGFW (next generation firewall)
OSPF (Open Shortest Path First)
overhead
point-to-point connection
PoP (point of presence)
prefix
prefix length
PSTN (public switched telephone network)
RIP (Routing Information Protocol)
RIPv2 (Routing Information Protocol, version 2)
route utility
routing cost
routing protocol
routing table
runt
SD-WAN (software-defined wide area network)
security group
stateful firewall
stateless firewall
static routing
symmetrical
UTM (unified threat management)

Acronyms Checklist

The acronyms in Table 9-4 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 9-4 Chapter 9 acronyms

Acronym	Spelled out
ACL	access control list
BPG	Border Gateway Protocol
CRC	Cyclic Redundancy Check
EIGRP	Enhanced Interior Gateway Routing Protocol
FHRP	First Hop Redundancy Protocol
IS-IS	Intermediate System to Intermediate System
OSPF	Open Shortest Path First
RIP	Routing Information Protocol
SD-WAN	software-defined wide area network
UTM	unified threat management

Review Questions

1. Which OSI layer is responsible for directing data from one LAN to another?
   a. Transport layer
   b. Network layer
   c. Data link layer
   d. Physical layer
2. When a router can't determine a path to a message's destination, where does it send the message?
   a. Default gateway
   b. Routing table
   c. Edge router
   d. Gateway of last resort
3. What virtual resource is used to protect network interfaces in the cloud?
   a. NACL
   b. Network security list
   c. Security group
   d. Virtual firewall
4. Which routing protocol does an edge router use to collect data to build its routing tables for paths across the Internet?
   a. RIPv2
   b. BGP
   c. OSPF
   d. IP
5. Which of the following ACL commands would permit web-browsing traffic from any IP address to any IP address?
   a. `access-list 101 deny tcp any any`
   b. `access-list 101 deny tcp host 2.2.2.2 host 3.3.3.3 eq www`
   c. `access-list 101 permit https any any`
   d. `access-list 101 permit icmp any any`
6. Who is responsible for the security of hardware on which a public cloud runs?
   a. The cloud customer
   b. It depends
   c. Both the cloud customer and the cloud provider
   d. The cloud provider
7. Which policy ensures messages are discarded when they don't match a specific firewall rule?
   a. Implicit allow
   b. Explicit deny
   c. Explicit allow
   d. Implicit deny
8. Which ACL rule will prevent pings from a host at 192.168.2.100?
   a. `access-list 101 permit icmp any host 192.168.2.100`
   b. `access-list 101 deny icmp host 192.168.2.100 any`
   c. `access-list 101 deny tcp host 192.168.2.100 host 192.168.2.1`
   d. `access-list 101 deny icmp any host 192.168.2.100`
9. What is the most likely cause of a firewall blocking traffic that should be allowed?
   a. Device failure
   b. Rule misconfigurations
   c. Rule exceptions
   d. Packet misconfiguration
10. When you list a router's routing table, one of the routes is labeled with an S. Which routing protocol was used to create this route?
    a. None
    b. OSPF
    c. EIGRP
    d. RIPv2

Scenario-Based Questions

Question 9-1

You're setting up a cloud deployment to run a business-critical database, along with backups and read-only copies distributed across multiple geographical regions to increase redundancy and decrease latency. Employees at your location must be able to access the data safely. Which cloud connectivity option will create a connection from your office network that is always available with industry-standard encryption for sensitive data in transit?

a. Direct connection
b. Internet
c. Client VPN
d. Site-to-site VPN

Question 9-2

You've just added a new network printer to the HR department's subnet, and you want to give Accounting department employees access to the printer as well. What kind of route is created when you configure the office router to use a specific path from the Accounting department subnet to the network printer?

a. Trace route
b. Static route
c. Default route
d. Dynamic route

Question 9-3

You're securing a network expansion for a high school, and school officials are concerned about the kinds of Internet access students might have on the network. Primarily, they want to ensure they can block traffic that contains problematic website content or gaming communications. What kind of firewall will give you the greatest ability to filter for this type of traffic?

a. Application layer firewall
b. Stateless firewall
c. Stateful firewall
d. Host-based firewall

Question 9-4

You're troubleshooting a problem on the network where an email server should be receiving traffic from the Internet, but the traffic is not reaching the server. You need to determine whether the router on the server's subnet has found the server to communicate with it. What command should you run on the router to determine if the router knows of the server's presence on the subnet?

a. `show interface`
b. `show route`
c. `show arp`
d. `show config`

Hands-On Projects

Note 9-9 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Project 9-1: Create WAN Links in Packet Tracer

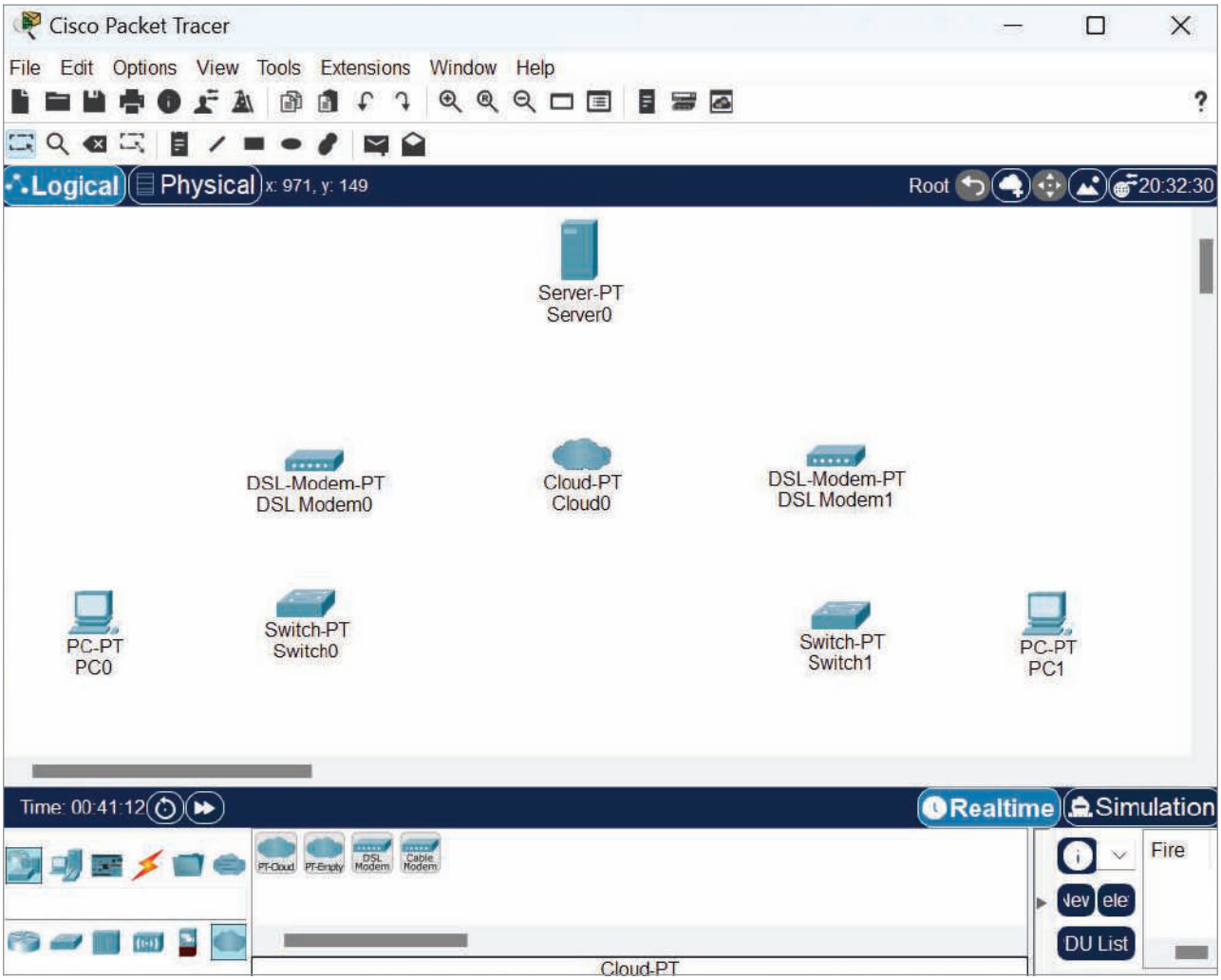
Estimated time: 20 minutes
Objective 1.3: Summarize cloud concepts and connectivity options.
Resources:
- Computer with Cisco Packet Tracer installed

Description:
This Packet Tracer project starts with a new network and is *not* intended to build on your Packet Tracer network for the Capstone Projects. For this project, do *not* use your existing Packet Tracer network (most recently worked on in Chapter 8)—you'll come back to that network file in this chapter's Capstone Projects. In this project, start with a new Packet Tracer file to experiment with WAN links. You will *not* need to save this network for future projects or chapters.

You can create several kinds of WAN connections in Packet Tracer. In this project, you'll keep the topology simple so you can see the process instead of getting mired in the details. You'll create DSL connections between two networks and a web server, which will host your own webpage. Complete the following steps:

1. Open Cisco Packet Tracer—make sure you're starting from a new file, not an existing network.
2. Add the following devices to your workspace (Figure 9-35):
   a. Two **PCs**
   b. One **Server**
   c. Two **PT-Switches**
   d. Two **DSL Modems** (from the Network Devices > WAN Emulation group)
   e. One **PT-Cloud** (from the Network Devices > WAN Emulation group)
3. Configure the Server with the following information:
   a. Static IP address: **192.168.2.10/24**
   b. Default Gateway: **192.168.2.1**
   c. DNS server: **192.168.2.10**
4. Configure the services you need on the Server. On the **Services** tab, click **DHCP**. Add the following information:
   a. DNS server: **192.168.2.10**
   b. Start IP address: **192.168.2.100**
   c. Subnet mask: **255.255.255.0**
   d. Maximum Number of Users: **50**
5. Click **Save** and then turn the DHCP service **On**.
6. Click **DNS**, configure the following information, click **Add** to create the new A record, and then turn **On** the DNS service:
   a. Name: **www.cengage.com**
   b. Address: **192.168.2.10**

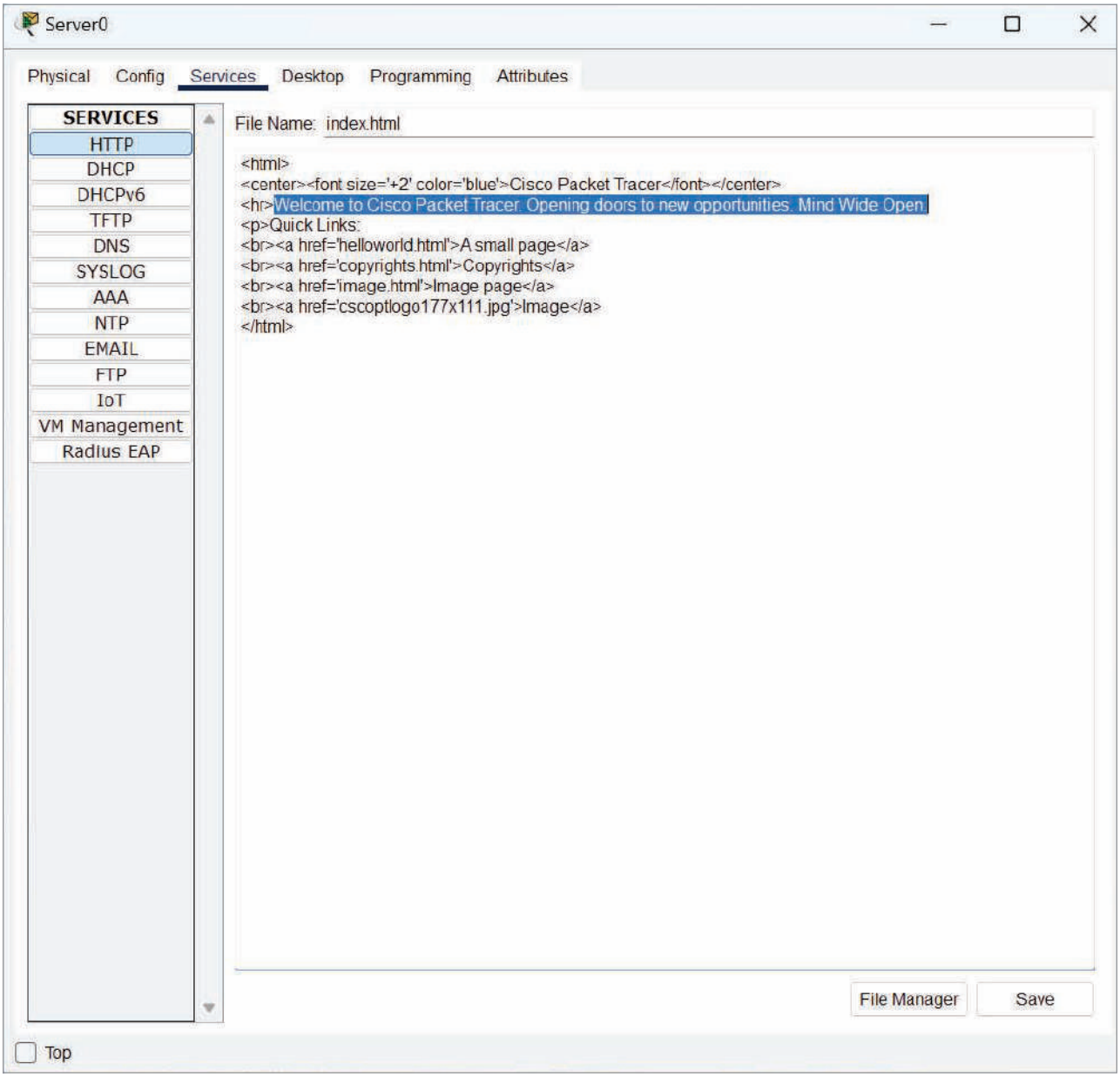
Figure 9-35 Add these devices to your network



Source: Cisco Systems, Inc.

7. Click **HTTP**. Next to the index.html file, click **(edit)**. As shown in Figure 9-36, delete the line that says "Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open." Replace that text with your own message, such as **Welcome to Jill West's webpage!!** (using your own name, of course). Click **Save** and then click **Yes**. Close the Server's configuration window.
8. Add and configure interfaces on the cloud. Click the **Cloud**. Turn off the physical device (scroll to the right to view the Power button if necessary). Drag one **PT-CLOUD-NM-1CFE** module to an available slot. Turn the physical device back on.
9. Click the **Config** tab and click **DSL**. At the top under DSL, make sure **Modem4** and **FastEthernet8** are selected for the ports and click **Add**. Then select **Modem5** and **FastEthernet8** and click **Add** again. Close the Cloud's configuration window.
10. Configure each PC to use **DHCP**.
11. Connect each PC to its Switch using a **Copper Straight-Through** cable.
12. Connect each Switch to its DSL Modem using a **Copper Cross-Over** cable. Use **FastEthernet** ports on the Switches, and **Port 1** on each Modem.
13. Connect each Modem to the Cloud using a **Phone** cable. On the Cloud, connect to the **Modem** ports.
14. Connect the Cloud (FastEthernet8) to the Server (FastEthernet0) using a **Copper Straight-Through** cable.

Figure 9-36 Replace the selected text with your own



Source: Cisco Systems, Inc.

15. Once all ports are up, wait a couple of minutes and then confirm that each PC received an IP address. What IP address did PC0 get? What about PC1?
16. On each PC, ping the Server. What command did you use? Does it work? If not, troubleshoot the problem.
17. On one PC, ping **www.cengage.com**. What command did you use? Does it work? If not, troubleshoot the problem.
18. On one PC, open a web browser and navigate to **www.cengage.com**.
19. **Take a screenshot** of the webpage showing your custom text; submit this visual with your responses to this project's questions.
20. Save your Packet Tracer network for future reference if desired. Make some notes on your Wikidot website about your activities in this project.

Note to instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site.

Project 9-2: Configure Cloud Security in AWS

Estimated time: 45 minutes (+10 minutes for group work, if assigned)
Objective 4.3: Given a scenario, apply network security features, defense techniques, and solutions.
Group work: This project includes enhancements when assigned as a group project.
Resources:
- AWS account created in Hands-on Project 7-1
- Internet access

Description:
In Chapter 7, you created an AWS account—either a standard account or an account through AWS Academy—and performed some tasks in the cloud. In this project, you will configure security resources in the cloud, including security groups and NACLs (network ACLs). Use the same AWS account you used in Chapter 7.

Caution | Depending on the status of your account and the selections you make during this project, an EC2 instance and its supporting resources (such as storage) can deplete your credits or accrue charges. Make sure to follow these steps carefully and delete all created resources at the end of this project.

Complete the following steps:

1. Sign into your AWS management console. If you're using an AWS Academy Learner Lab, you'll need to access your AWS management console through your AWS Academy course. If you're using a standard AWS account, sign in directly at **aws.amazon.com**.
2. Go to the **EC2** dashboard and click the orange **Launch instance** button.
3. Give your instance a name, such as Proj9-2. Select the free tier eligible **Amazon Linux 2023 AMI**.
4. Make the following selections:
   a. Keep the default selection of the free tier eligible **t2.micro** instance type.
   b. For Key pair name, choose **Proceed without a key pair (Not recommended)**. You will not need a key pair for this project.
   c. In the Network settings section, click **Edit**.

Security groups provide a secure perimeter around each network interface on an EC2 instance. Each interface can be assigned up to five security groups. A security group blocks all traffic and will only allow traffic that is explicitly allowed. The default selection is to create a rule that will allow SSH from anywhere (0.0.0.0/0) into your Linux EC2 instance through port 22. To explore some of your options here, complete the following steps:

5. Do not change the rule type, but click the **Type** arrow to see other options. What are three other protocols you could choose for rules in your security group?
6. What happens when you change the Source type to **My IP**? (Be sure to save the displayed information for the group assignment later in this project.)
7. Change the Source back to **Anywhere**, don't make any additional configuration changes, and launch your instance.

Now you're ready to connect with your instance:

8. Once it's running, select your instance and click **Connect**.
9. On the EC2 Instance Connect tab, click **Connect**.

Now that you're connected with your instance, let's experiment with how well the instance's security group works:

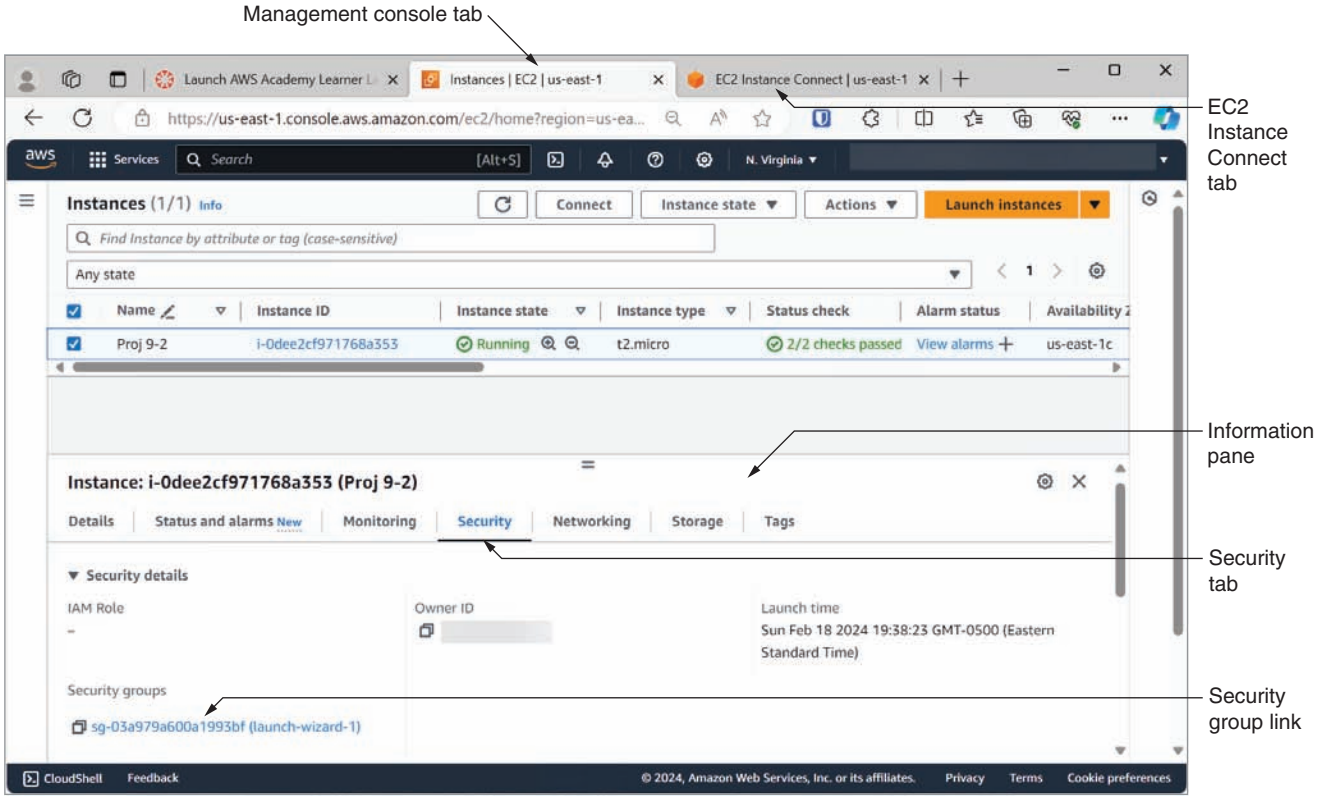
10. When the connection is established, try to ping Google's DNS server at 8.8.8.8. Did it work? Why do you think this is?

11. Recall that an EC2 instance's security group is stateful, meaning it will allow returning traffic for a conversation that was initiated by the EC2 instance itself, such as a ping. Check your management console tab for information about your instance. What is this instance's public IP address?
12. Try pinging this instance from your local computer. Does it work? Why do you think this is?

Let's change this. To add a new rule to your instance's security group, complete the following steps:

13. On the management console tab in your instance's information pane, click the **Security** tab (Figure 9-37). Scroll down to the Security groups section and click the security group's link.

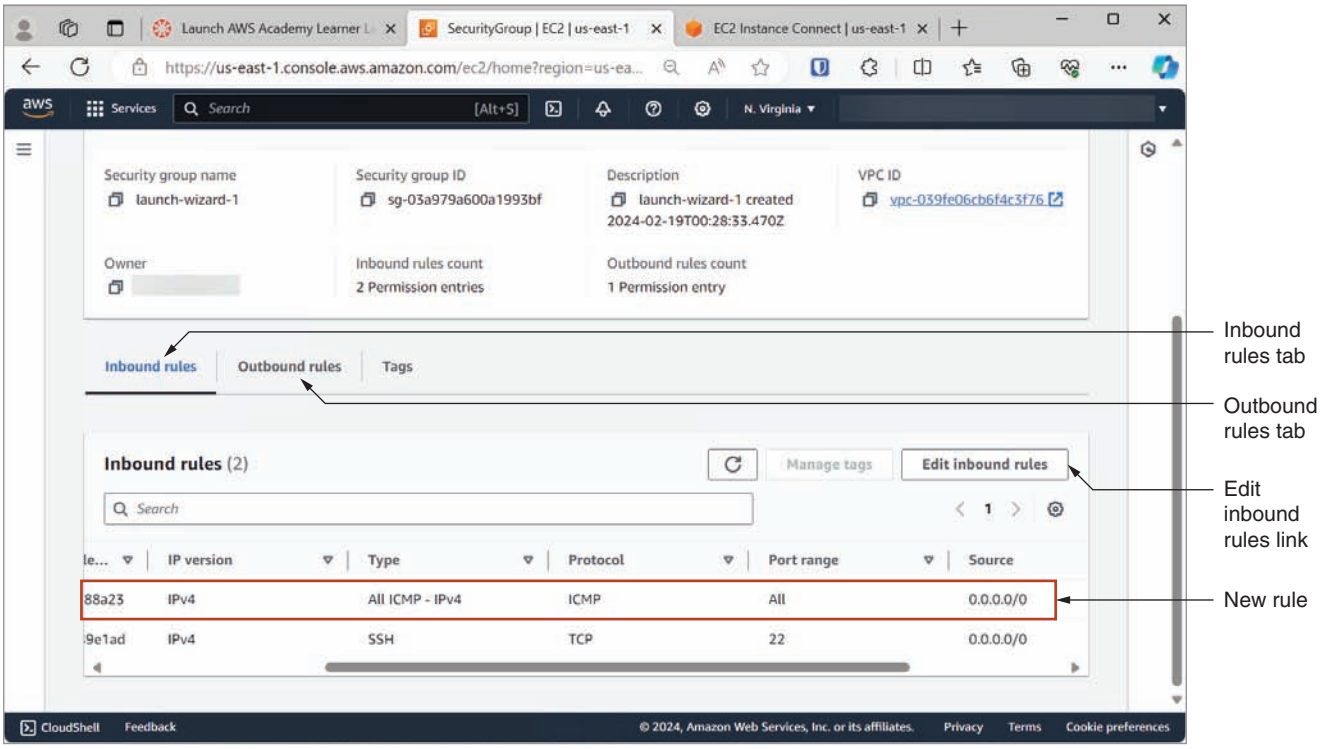
Figure 9-37 Security groups are listed on the Security tab



Source: Amazon Web Services, Inc.

14. Explore the **Inbound rules** tab and the **Outbound rules** tab, and respond to the following questions:
    a. How many inbound rules are currently listed?
    b. What traffic is currently allowed to enter the security group, and from what source?
    c. How many outbound rules are currently listed?
    d. What traffic is currently allowed to exit the security group, and to what destination?
15. On the **Inbound rules** tab, click **Edit inbound rules**. Add a new rule that allows **All ICMP - IPv4** traffic from any IPv4 address. Save the rules.
16. You should now have a new rule that allows ICMP traffic on any port from any source (Figure 9-38).
17. Try again to ping your EC2 instance from your local computer. Does it work? Why do you think this is?

Figure 9-38 New rule allows incoming ping traffic



Source: Amazon Web Services, Inc.

By default, your VPC's NACL allows all traffic. You can add allow or deny rules to further control traffic into and out of each subnet. Recall that NACL rules are stateless, meaning a protocol's traffic must be allowed in both directions for a conversation such as ping to succeed. Complete the following steps:

18. Return to your instance's information pane and click the **Networking** tab. Locate and click the instance's VPC ID. This takes you to your VPC in the VPC dashboard.
19. Select the VPC listed here (there should just be one). In the VPC's information pane, click the link for the **Main network ACL**. This takes you to the list of NACLs.
20. Select the NACL listed here (there should just be one). In the NACL's information pane, click the **Inbound rules** tab. What inbound rules are listed here?

Notice that these rules seem contradictory—why would one rule allow all traffic and the next rule deny all traffic? Each rule is assigned a priority. The NACL checks each rule in succession from highest priority to lowest priority. As soon as a rule is found that matches the traffic, that rule is applied and no additional rules are evaluated. To block ICMP traffic (which is currently allowed by both the NACL and the security group), complete the following steps:

21. Click **Edit inbound rules** and then click **Add new rule**.
22. The new rule will need to be a higher priority than the existing "allow all traffic" rule that is rated at 100. Choose a number below 100 for your new rule, such as 90.
23. Continue configuring the rule so it will deny all ICMP - IPv4 traffic from all sources. Be sure to save your changes.
24. Display a list of all three inbound rules for the NACL. **Take a screenshot** showing your new NACL inbound rule; submit this visual with your responses to this project's questions.
25. If your goal is only to block outside computers from pinging your EC2 instance, is this NACL inbound rule sufficient to accomplish your goal, or do you also need a corresponding outbound rule?
26. Try again to ping your EC2 instance from your local computer. Does it work? Why do you think this is?

27. **For group assignments:** Currently, you can't ping your instance from your local computer due to the NACL rule blocking incoming ICMP traffic. This rule applies to all source IP addresses. How can you configure your NACL to allow pings from your computer but still block pings from other computers on the Internet? Each partner should configure a new, higher priority NACL rule that allows ICMP traffic only from their own public IP address. Confirm you can now ping your instance. Next, give a partner your instance's public IP address, and then try to ping your partner's instance. Does it work? Why do you think this is?
28. Return to your list of instances in EC2, and close the EC2 Instance Connect tab and the Network ACLs tab.
29. Recall that turning off the instance does not delete any resources in your AWS cloud. In some cases, you continue to be charged for resources even if they're turned off. To delete your instance, select the instance, click **Instance state**, and click **Terminate instance**. Click **Terminate**. What is the instance state now?
30. If you used AWS Academy for this project, be sure to end the lab session in the AWS Academy console.

Caution | Depending on the status of your account and the selections you made during this AWS project, your EC2 instance and its supporting resources (such as storage) can deplete your credits or accrue charges. Double-check to make sure you've terminated all resources you created in this project.

Capstone Projects

Note 9-10 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Capstone Project 9-1: Configure RIP on Routers in Packet Tracer

Estimated time: 30 minutes
Objective 2.1: Explain characteristics of routing technologies.
Resources:
- Computer with Cisco Packet Tracer installed
- Access to the Packet Tracer network created in Capstone Project 8-2
- Storage space for Packet Tracer network file to be accessed in later projects and chapters

Description:
This Capstone Project picks up where Capstone Project 8-2 left off. Currently, your PCs can't communicate beyond their own LAN. A routing protocol will enable connections across multiple LANs. In this project, you'll configure dynamic routing across your network using RIPv2. Although RIPv2 is an outdated routing protocol, it's easier to configure than more modern routing protocols and will therefore give you a basic introduction to how these protocols work. Complete the following steps:

1. In **Cisco Packet Tracer**, open your Packet Tracer file from Capstone Project 8-2. If desired, make a backup copy before proceeding.

2. Determine which PCs can talk to each other. Test the following pings:
   a. From PC0, ping PC1. Does it work?
   b. From PC0, ping any of Router3's interfaces. Does it work?
   c. From PC0, ping any of Router2's interfaces. Does it work?
   d. From PC0, ping PC6. Does it work?
3. **Take a screenshot** of your Packet Tracer network and mark all the devices PC0 can successfully ping; submit this visual with your responses to this project's questions.

Now you're ready to enable dynamic routing on your network. Complete the following steps:

4. Click **Router0**. In Router0's CLI, enter the commands listed in Table 9-5.

Table 9-5 Configure RIPv2 on each router

Command	Purpose
`enable`	Enters privileged EXEC mode
`configure terminal`	Enters global configuration mode
`router rip`	Enables RIP
`version 2`	Specifies RIPv2, which is required to support classless routing
`network 192.168.43.0`	Specifies connected network to advertise—all networks directly connected to the router must be listed to be included in RIPv2 routing tables
`network 192.168.43.16`	Specifies connected network to advertise
`network 192.168.43.24`	Specifies connected network to advertise
`network 192.168.43.32`	Specifies connected network to advertise
`exit`	Returns to global configuration mode
`exit` and press **Enter**	Returns to privileged EXEC mode
`show ip route rip`	Displays routing table entries learned through RIP
`copy run start` and press **Enter**	Saves the current settings

Note 9-11 | Depending on the order in which you enable routing on the routers, each router's routing table might initially be empty. Dynamic routing must be enabled on multiple connected routers before routers can begin to build their routing tables.

5. Repeat the commands listed in Table 9-5 for each of the routers, substituting the network addresses directly connected to each router, as follows:
   a. **Router1**—List the network addresses **192.168.43.4**, **192.168.43.48**, and **192.168.43.64**.
   b. **Router2**—List the network addresses **192.168.43.8**, **192.168.43.80**, and **192.168.43.96**.
   c. **Router3**—List the network addresses **192.168.43.0**, **192.168.43.4**, and **192.168.43.8**.

Note 9-12 | Note that Router0 is connected to four subnets because it also supports two VLANs, and the other routers are each connected to three subnets.

6. As RIP is enabled on each router, the router learns of the routes connected to it. To see the routes learned by Router3, enter the command **`show ip route rip`**. How many routes has Router3 learned through RIP?
7. In addition to learning connected routes, the router also collects MAC addresses for interfaces communicating with each of its own interfaces on local subnets. The router maps these MAC addresses to IP addresses in its ARP table. To see the interfaces Router0 is aware of, enter the command **`show arp`**. How many interfaces are listed in Router0's ARP table? This information will change after the next steps.
8. Determine which PCs can talk to each other. Test the following pings:
   a. From PC0, ping PC1. Does it work?
   b. From PC0, ping any of Router3's interfaces. Does it work?
   c. From PC0, ping any of Router2's interfaces. Does it work?
   d. From PC0, ping PC6. Does it work?
9. **Take a screenshot** of your Packet Tracer network and mark all the devices PC0 can successfully ping; submit this visual with your responses to this project's questions.
10. Return to Router0 and enter the **`show arp`** command again. How many interfaces are listed in its ARP table now?
11. Save this Packet Tracer file in a safe place for future projects.
12. Make some notes on your Wikidot website about your activities in **Packet Tracer** for this project. When you're finished, close Packet Tracer or continue to Capstone Project 9-2.

Note to instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site. Some Packet Tracer projects build on earlier Packet Tracer networks. If needed for one or more students, you can provide a previous project's solution file as a start file for one of these progression projects.

Capstone Project 9-2: Sabotage and Repair Interfaces in Your Packet Tracer Network

Estimated time: 45 minutes (+20 minutes for group work, if assigned)
Objective 5.3: Given a scenario, troubleshoot common issues with network services.
Group work: This project includes enhancements when assigned as a group project.
Resources:
- Computer with Cisco Packet Tracer installed
- Access to the Packet Tracer network created in Capstone Project 9-1
- Storage space for Packet Tracer network file to be accessed in later projects and chapters

Description:
This Capstone Project picks up where Capstone Project 9-1 left off. In this project, you'll experiment with commands covered in this chapter to troubleshoot network problems. The commands covered in this chapter are intended to help you identify and correct device configuration problems. To do this, you'll need to remember what commands are available to you and understand how and when to use them. Complete the following steps:

1. In **Cisco Packet Tracer**, open your Packet Tracer file from Capstone Project 9-1.
2. Take a few moments to document your current configurations. For example, test your pings and make sure everything works. Make sure that all your notes in your workspace match the actual configurations of your switches, routers, and PCs. Take screenshots or notes about all these configurations so this information is recorded somewhere else in addition to being visible on your network.

3. Save a copy of your Packet Tracer file so you'll have a backup file you can check as changes are made to your test file. What is the name of your backup file?
4. Table 9-6 lists a variety of ways you might create a problem on your network. Make one change at a time to sabotage your network. For each problem, document the following information:
   a. **Symptoms**—Identify symptoms of the problem. Ask what problems network users might notice or what indications would identify that a problem exists.
   b. **Tests**—List any tests you performed to identify and confirm the problem. For example, you could use the `show vlan` command to help you identify a missing or misconfigured VLAN, or use the `show ip route` command to help you identify a missing or misconfigured route.
   c. **Solution**—Repair the problem and confirm the network is functioning as expected. How did you repair the problem?
   d. Repeat this process for each problem listed in Table 9-6. Complete the table as you go.

Table 9-6 Network problems and solutions

Problem	Symptoms	Tests	Solution
Change IP address			
Change subnet mask			
Change default gateway			
Change VLAN			
Change port mode (i.e., access or trunk)			
Delete VLAN			
Shut down port			
Require DHCP with no DHCP server			
Disable RIP			
Move cable to wrong port			
Remove physical module			
Power down device			

5. When you're finished, **take a screenshot of your completed table**; submit this visual with your responses to this project's questions.
6. **For group assignments:** Trade computers with a team member. Using Table 9-6 as a reference, choose one problem and sabotage your teammate's network. Then go back to your own network and identify the problem your teammate instituted. Document the symptoms and the tests you used to identify the problem, and then fix the problem. What problem did you find and how did you fix it? Repeat this process for two more problems.
7. Retest your network to ensure all PCs can ping each other. Confirm all devices are configured correctly. As needed, refer to your notes from Step 2.

8. Save this Packet Tracer file in a safe place for future projects.
9. Make some notes on your Wikidot website about your activities in Packet Tracer for this project. When you're finished, close **Packet Tracer**.

Note to instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site. Some Packet Tracer projects build on earlier Packet Tracer networks. If needed for one or more students, you can provide a previous project's solution file as a start file for one of these progression projects.

Solutions to Self-Check Questions

Section 9-1: WAN Connectivity

1. You just moved into a rural office space that has telephone service but no cable. Which WAN service could you use without needing to install new wiring to your location?
   Answer: b. DSL
   Explanation: DSL (digital subscriber line) operates over the PSTN (public switched telephone network), which is a network of lines and carrier equipment that provide landline telephone service to homes and businesses. Cable broadband uses the coaxial cable wiring used for TV signals. Fiber broadband and leased fiber both require fiber cables, although it's possible there could be a fiber broadband service that reaches a nearby node, which then runs over the phone lines for the final stretch to the office's location.

2. Which of these WAN services is backed by an SLA?
   Answer: b. Leased line
   Explanation: Leased line performance is backed by SLA-enforced uptime, repair time, and possibly backup options. DSL, cable, and fiber broadband all rely on a "best-effort" service level to provide *up to* the advertised bandwidth, and actual performance varies considerably during busy usage.

3. Which WAN service offers active-active load balancing?
   Answer: c. SD-WAN
   Explanation: An **SD-WAN (software-defined wide area network)** managed network offers active-active load balancing where it can choose the best physical WAN connection for different types of traffic according to traffic prioritization and current network conditions. DSL, cable, and fiber broadband all offer "best-effort" service over a single connection to the ISP.

Section 9-2: Routing Protocols

4. Which routing protocol runs between your network's edge router and your ISP's edge router?
   Answer: d. BGP
   Explanation: BGP (Border Gateway Protocol) spans multiple autonomous systems and is used by edge and exterior routers on the Internet. IGPs, including EIGRP, RIPv2, and OSPF, are used by core routers and edge routers within autonomous systems.

5. Which command will output your Windows computer's routing table?
   Answer: b. `route print`
   Explanation: The **`route print`** command can be used to view a Windows computer's routing table and to add or delete static routes. On Linux or UNIX systems, use the command `route`. On a Cisco device, use the command `show ip route`. The `vrrp` command is used to configure VRRP (Virtual Router Redundancy Protocol) on routers.

6. Which routing protocol is limited to 15 hops?
Answer: d. RIPv2
Explanation: The 15-hop limit is specific to RIP and **RIPv2**. This is an identifying factor you can use to distinguish RIP from other routing protocols, such as BGP, OSPF, and EIGRP.

Section 9-3: Securing WAN Connections

7. Which device can be used to increase network performance by caching websites?
Answer: b. Proxy server
Explanation: A **proxy server** can improve performance for users accessing resources external to their network by caching files. A firewall is a specialized device or software that selectively filters or blocks traffic between networks. An IDS (intrusion detection system) monitors network traffic, generating alerts about suspicious activity. An AWS security group filters traffic into and out of a single EC2 instance.

8. Which firewall type can protect a home network from adult content not suitable for the family's children?
Answer: d. Application layer firewall
Explanation: Application layer firewalls can block designated types of traffic based on application data contained within packets. The simplest form of a firewall is a packet-filtering firewall, which is a network device or application that only examines each message's header. Host-based firewall software only protects the computer on which it's installed. A stateless firewall manages each incoming packet as a stand-alone entity without regard to currently active connections.

Section 9-4: Troubleshooting WAN Connections

9. Where is a router's hostname stored when you first change the name?
Answer: d. Running-config file
Explanation: When making a configuration change to a Cisco device such as changing its hostname, those changes are held in the **running-config file**. To make your changes persist beyond a device's power cycle, you must copy the running-config file to the startup-config file. A routing table is a database that holds information about where hosts are located and the most efficient way to reach them. A MAC address table is used by a switch to match MAC addresses with physical ports on the switch.

10. Which problem is most likely caused by a damaged cable?
Answer: b. CRC error
Explanation: A **CRC (Cyclic Redundancy Check) error** indicates messages are being damaged in transit, such as when there's a cable problem or a damaged NIC. A routing loop can be caused by making too many topology changes too quickly. Asymmetrical routing is caused when messages going in one direction in a conversation (such as from a web server to a client) travel a different path than messages going in the other direction (such as from client to web server). Giants are frames that are too large and they are dropped; excessive giants being reported is usually a result of misconfigurations.